Anirudh Jayaprakash

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Professional Summary

Machine Learning Engineer with hands-on experience building diffusion models, multi-agent AI systems (CrewAI, LangChain), and production ML pipelines. Published researcher (IEEE Xplore, GCAT 2023) with expertise in computer vision, NLP, and GenAI. Proficient in Python, PyTorch, TensorFlow, SQL, and cloud-native deployment using Docker, AWS, and CI/CD.

Technical Skills

Languages: Python, C, C++, Java, SQL, R.
Machine Learning & AI: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, Diffusion Models, U-Net, Swin Transformers ,vLLM, MCP, Generative AI.
LLM & Agentic AI: RAG, Prompt Engineering, LangChain, CrewAI, Vector Embeddings, LLM Fine-Tuning (LoRA/QLoRA).
Data Engineering & Big Data: ETL/ELT Pipelines, Distributed Data Processing, Apache Spark, Apache Airflow, Snowflake, Databricks, Data Lakes, Data Validation.
Cloud & MLOps: AWS, Azure, Docker, Jenkins, CI/CD, MLflow, Git.
Web, APIs & Databases: Flask, Gradio, REST APIs, Selenium, PostgreSQL, MySQL, MongoDB, SQLite, pgvector, ChromaDB.

Projects

Multi-LoRA Adapter Serving with vLLM | *GitHub* — August 2026 – September 2026
- Fine-tuned 3 task-specialized **LoRA adapters** (SQL generation, dialogue summarization, NER-to-JSON extraction) on a shared **Llama-3.2-3B** base using PEFT/TRL, then served all 3 **simultaneously from a single vLLM engine** via **LoRARequest** routing, eliminating reloading and duplicate model copies.
- Debugged and resolved **5+** compatibility issues across the **PyTorch/vLLM/trl** stack (GPU kernel mismatches, KV-cache memory limits, deprecated dataset loading, library API drift), documenting root causes and fixes for reproducibility.
- Demonstrated measurable behavioral gains from task-specific adapters over the generic base model, e.g., the extraction adapter emitted clean, directly parseable JSON where the base model wrapped output in markdown and prose that broke naive parsing.

Diffusion-Based Image Super-Resolution with ResShift | *GitHub* — November 2025 – January 2026
- Engineered a diffusion-based super-resolution architecture with **U-Net** and **Swin Transformer** blocks for 4x image enhancement.
- Integrated **VQGAN latent-space processing** to improve generation quality and computational efficiency.

NBA Insights API — Multi-Agent LLM System | *GitHub* — October 2025 – December 2025
- Built a multi-agent NBA Q&A system with **CrewAI**, FastAPI, and Pandas, exposing a REST API with 5 structured data tools; achieved **100% tool-use rate** on a golden eval set.
- Added reliability guardrails to the agent pipeline (groundedness gate, **TTL caching**, retry logic, tool-call budgeting), reaching 1.0 faithfulness and **92% tool-call success at 2.7s p50 latency.**

Experience

University of Maryland, College Park — September 2025 – May 2026
Graduate Assistant — *College Park, MD*
- Built **Retrieval-Augmented Generation** (RAG) pipelines using **LangChain** and **CrewAI** with OpenAI API integration to automate literature and data analysis tasks for research projects.
- Built and benchmarked a local LLM inference stack (Ollama, FastAPI) for the research team, evaluating models on time-to-first-token, decode throughput, and latency, identified up to **6x** throughput differences across model choices to inform tooling decisions.
- Performed statistical analysis and hypothesis testing (Python: pandas, SciPy/statsmodels; R) on research datasets, building visualizations and dashboards to communicate findings to the research team.

Urban Ventures — October 2023 – March 2024
Web Developer Intern — *India*
- Developed responsive web interfaces using JavaScript, jQuery, and **Bootstrap**, increasing user engagement by **30%**.
- Integrated **REST APIs** and optimized mobile rendering, reducing bounce rate by **25%**.

lastminute.com — February 2023 – August 2023
Software Developer Intern — *India*
- Automated **QA** workflows through modular **Selenium** and Appian test suites, reducing manual QA effort by **45%**.
- Refactored legacy tests into data-driven scripts, improving runtime efficiency by **30%**.
- Validated REST APIs and supported **Jenkins CI/CD** pipelines to accelerate release cycles.

Education

University of Maryland, College Park — August 2024 – May 2026
M.S. in Data Science — *College Park, MD*
CMR Institute of Technology (CMRIT) — 2019 – 2023
B.E. in Information Science — *Bangalore, India*

Publications

A. Jayaprakash et al., *"Knowledge Graph Based Medical Chatbot Building,"* in Proc. 2023 4th IEEE Global Conference for Advancement in Technology (GCAT), 2023. View on IEEE Xplore